

BRAC University
Dept. of Computer Science and Engineering
Spring 2025
CSE331-02

Time: 20 minutes

Marks: 20

Name:

ID:

Section:

1. Convert the following regular expression into context free grammar. [8]

$$a + (cc+ab)^*cb + (c+ab)^*b$$

2. Give a context-free grammar that generates the language [6+6=12]

a. $L = \{w \in \{a, b, c\}^* : w = b^i a^j c^k, \text{ where } k \geq i + 2j, i, j \geq 0\}$

b. $L = \{w \in \{0, 1\}^* : w \text{ contains 1 only at odd positions}\}$

1. $S \rightarrow a \mid x \mid Y$

$$X \rightarrow PQ$$

$$P \rightarrow RP \mid \epsilon$$

$$R \rightarrow cc \mid b$$

$$Q \rightarrow cb$$

$$Y \rightarrow MN$$

$$M \rightarrow c \mid D$$

$$D \rightarrow bD \mid \epsilon$$

$$N \rightarrow b$$

2. a) $k = i + 2j + D \rightarrow 0 + 0 + \infty$

$$\frac{b^i a^j c^{2j} c^i c^D}{x \quad y}$$

$$S \rightarrow XY$$

$$X \rightarrow bXc \mid A$$

$$A \rightarrow aAec \mid \epsilon$$

$$Y \rightarrow cY \mid \epsilon$$

b) $(0+1)0^*(\epsilon+0+1)$

$$S \rightarrow 00S \mid 10S \mid \epsilon \mid 0 \mid 1$$